

D-MODULES AND MIXED HODGE THEORY

1. INTRODUCTION

During the whole of these notes k will denote a field of characteristic 0. All of the algebras here are assumed to be unital, but not necessarily commutative.

These notes may lack exercises. If there is a need for them, the author proposes to prove some of the lemmas, propositions or theorems by your own.

2. THE FILTERED ALGEBRA OF DIFFERENTIAL OPERATORS OF A COMMUTATIVE ALGEBRA

During this section R will denote a commutative k -algebra.

2.1. Basic definitions and properties.

Observe that the algebra of k -linear endomorphism¹ $End_k(R)$ is a noncommutative k -algebra; therefore, in particular, a Poisson algebra. Our first task is to recursively define differentials operators of order n on R , where n is a non-negative integer. The collection of all operators of finite order, denoted by $D(R)$ will be graded sub k -algebra of $End_k(R)$ called the algebra of differential operators of R .

One warning about notation abuse: for a given $a \in R$, we will write a both for the element of R and the element of $End_k(R)$ given by multiplication by a . The difference is relevant, and we should be aware of the context.

Definition 2.1. Let $P \in End_k(R)$. P is called a **differential operator (on R) of order zero** if $[P, a] = 0$ for every $a \in R$. Now, suppose $n > 0$. P is called a **differential operator (on R) of order n** if P is not a differential operator of order less than n , and $[P, a]$ is a differential operator of order less than n for every $a \in R$. Let us denote by $D^n(R)$ the set of all differential operators of order less than or equal to n . We also define $D(R) = \bigcup_{n < \omega} D^n(R)$.

Proposition 2.2. (1) Let $P \in End_k(R)$, and $n < \omega$. Then, $P \in D^n(R)$ if, and only if,

$$[[\cdots [P, a_0] \cdots], a_{n-1}], a_n] = 0$$

for every $a_0, \dots, a_n \in R$.

(2) Let $P \in End_k(R)$, and $n < \omega$. Then, $ord(P) = n$ if, and only if,

$$[[\cdots [P, a_0] \cdots], a_{n-1}], a_n] = 0$$

for every $a_0, \dots, a_n \in R$, and

$$[\cdots [P, b_0] \cdots], b_{n-1}] \neq 0$$

for some $b_0, \dots, b_{n-1} \in R$.

(3) $D^m(R) + D^n(R) \subset D^{\max\{m, n\}}(R)$.

(4) $D^m(R) \cdot D^n(R) \subset D^{m+n}(R)$.

(5) $D^0(R) = R$.

(6) $D^1(R) = R + Der_k(R)$.

¹Not necessarily preserving the ring product.

Proof. (1) Our proof will be by induction on n . Observe that the base case $n = 0$ is just the definition of differential operators of order 0. Let us assume that the proposition holds for n . Let $P \in D^{n+1}(R)$. We have two cases. For the first case, suppose $P \in D^n(R)$, then, by the induction hypothesis,

$$[[\cdots [P, a_0] \cdots], a_{n-1}], a_n] = 0$$

for every $a_0, \dots, a_{n+1} \in R$. Therefore

$$[[[[\cdots [P, a_0] \cdots], a_{n-1}], a_n], a_{n+1}] = 0$$

for every $a_0, \dots, a_n, a_{n+1} \in R$. For the second case, $P \notin D^n(R)$. Therefore P has order $n + 1$. Let $a_0, \dots, a_n \in R$ be arbitrary. By definition of differential operator of order $n + 1$, we have that $[P, a_0] \in D^n(R)$. The induction hypothesis yields

$$[[[[\cdots [P, a_0] \cdots], a_{n-1}], a_n], a_{n+1}] = 0$$

for every $a_0, \dots, a_n, a_{n+1} \in R$. This concludes one direction of the induction step.

For the reverse direction, suppose that

$$[[[[\cdots [P, a_0] \cdots], a_{n-1}], a_n], a_{n+1}] = 0$$

for every $a_0, \dots, a_n, a_{n+1} \in R$. Observe that this implies that $[P, a] \in D^n(R)$ for every $a \in R$. Here we have two cases. If $P \in D^n(R)$, then $P \in D^{n+1}(R)$. On the other hand $P \notin D^n(R)$. Then, by definition, P is of order $n + 1$, which implies $P \in D^{n+1}(R)$. This concludes our inductive step.

- (2) This is a similar proof to the previous one.
- (3) Immediate consequence of (1).
- (4) We are going to prove this statement in two different ways. The first one uses induction, and the second one is by means of explicit computations. Both rely on the fact that $\text{End}_k(R)$ is a Poisson algebra; which implies, in particular, that for every $P, Q \in \text{End}_k(R)$ and $a \in R$, we have

$$[PQ, a] = [P, a]Q + P[Q, a]. \quad (2.1)$$

In other words, $[\cdot, a]$ is a derivation of $\text{End}_k(R)$. The first proof is done by induction on $l = m + n$. For $l = 0$, we have $m = 0$ and $n = 0$, and $D^0(R) \cdot D^0(R) \subset D^0(R)$ follows from equation (2.1). Assume that the statement holds for $l > 0$. Let $l + 1 = m + n$, and $P \in D^m(R)$, $Q \in D^n(R)$. An easy consequence of the definition of differential operator is that $[P, a] \in D^{m-1}(R)$ and $[Q, a] \in D^{n-1}(R)$ for every $a \in R$. Then, by the induction hypothesis, $[P, a]Q, P[Q, a] \in D^{m+n-1}(R)$. This, together with equation (2.1) and the previous item of this proposition, implies that $[PQ, a] \in D^{m+n-1}(R)$. Here we have two cases: $PQ \in D^{m+n-1}(R)$, we trivially conclude the induction step; on the other hand, if $PQ \notin D^{m+n-1}(R)$, then, by definition, we conclude that PQ has order $m + n$, which implies $PQ \in D^{m+n}(R)$. Therefore, the induction step is completed.

- (5) Since R is a commutative, every multiplication by an element of R defines a differential operator of order 0. Now, let $P \in D^0(R)$. Then, for every $a \in R$, $0 = [P, a](1) = P(a) - aP(1)$. This implies that P is the multiplication by $P(1)$.
- (6) It is easy to see that every non-zero k -derivation of R is a differential operator of order 1. Therefore, $R + \text{Der}_k(R) \subset D^1(R)$. Let $P \in D^1(R)$. Then, for any a we have that $[P, a]$ is the multiplication by $P(a) + aP(1)$. Let $b \in R$. Therefore $P(ab) - aP(b) = [P, a](b) = bP(a) + abP(1)$. Which implies $P - P(1) \in \text{Der}_k(R)$.

□

Observe that the previous proposition implies that $D(R)$ is a graded, noncommutative R -algebra, and, in particular, a graded Poisson k -algebra.

2.2. The Weyl algebra is the algebra of differential operators on a polynomial algebra.

This section is devoted to giving an explicit description of $D(k[x_1, \dots, x_n])$.

First, for $i \in \{1, \dots, n\}$, let us denote by $\partial_i \in \text{Der}_k(k[x_1, \dots, x_n])$ the (formal) partial derivative $\partial_i(f) := \frac{\partial f}{\partial x_i}$. For any tuple $\alpha = (\alpha_1, \dots, \alpha_n) \in \omega^n$, we define

$$\begin{aligned} |\alpha| &= \alpha_1 + \dots + \alpha_n; \\ \alpha! &= \alpha_1! \cdots \alpha_n!; \\ \partial^\alpha &= \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n}; \\ x^\beta &= x_1^{\beta_1} \cdots x_n^{\beta_n}. \end{aligned}$$

We have the following commutation relations in $\text{End}_k(k[x_1, \dots, x_n])$:

$$\begin{aligned} [x_i, x_j] &= [\partial_i, \partial_j] = 0 & \text{for any } i, j \in \{1, \dots, n\}; \\ [\partial_i, x_j] &= \delta_{ij} & \text{for any } i, j \in \{1, \dots, n\}. \end{aligned}$$

Definition 2.3. The n th-Weyl algebra over k , denoted by $A_n(k)$, is the k -subalgebra of $\text{End}_k(k[x_1, \dots, x_n])$ generated by x_i and ∂_i for $i = 1, \dots, n$.

Lemma 2.4. Let $\alpha, \beta \in \omega^n$ such that $|\beta| \leq |\alpha|$. Then $\partial^\alpha(x^\beta) = \alpha! \cdot \delta_{\alpha\beta}$.

Proof. This follows from $\partial^\alpha(x^\beta) = \partial_1^{\alpha_1}(x_1^{\beta_1}) \cdots \partial_n^{\alpha_n}(x_n^{\beta_n})$. $\square$

Proposition 2.5. The set $\{\partial^\alpha \mid \alpha \in \omega^n\}$ is a basis for $A_n(k)$ as a left $k[x_1, \dots, x_n]$ -module. The set $\{x^\alpha \partial^\beta \mid \alpha, \beta \in \omega^n\}$ is a basis for $A_n(k)$ as a k -vector space.

Proof. The commutation relations imply the generating condition. For linear independence, let $\sum p_\beta \partial^\beta = 0$ be a finite sum with $p_\beta \in k[x_1, \dots, x_n]$, and let $\alpha \in \omega^n$ be an index of maximal modulus in the previous sum. Then, by the previous lemma, $\alpha! p_\alpha = (\sum p_\beta \partial^\beta)(x^\alpha) = 0$. Which implies $p_\alpha = 0$. In this way we obtain $p_\beta = 0$ for every β . $\square$

The previous proposition allows us to define the order of an operator $P \in A_n(k)$ in the following way: P can be written uniquely in the form $\sum_{\alpha \in \omega^n} p_\alpha \partial^\alpha$, where $p_\alpha \in k[x_1, \dots, x_n]$, and $p_\alpha \neq 0$ only for a finite number of $\alpha \in \omega^n$. We define the **order** of P as the maximum of $|\alpha|$ where α is such that $p_\alpha \neq 0$. The order of $P = 0$ is defined to be 0.

Proposition 2.6. $\text{Der}_k(k[x_1, \dots, x_n])$ is the $k[x_1, \dots, x_n]$ -submodule of $\text{End}_k(k[x_1, \dots, x_n])$ generated by $\partial_1, \dots, \partial_n$.

Proof. $\square$

Proposition 2.7. $A_n(k) \subset D(k[x_1, \dots, x_n])$.

Proof. Any element of $A_n(k)$ can be written in the form $\sum_{\alpha \in \omega^n} p_\alpha \partial^\alpha$, where $p_\alpha \in k[x_1, \dots, x_n]$, and $p_\alpha \neq 0$ only for a finite number of $\alpha \in \omega^n$. Then, by Proposition 2.2 (3)-(6), and Proposition 2.6, we have that $\sum_{\alpha \in \omega^n} p_\alpha \partial^\alpha$ belongs to $D(k[x_1, \dots, x_n])$. $\square$

Our aim in this subsection was already announced at its title. It is to prove that, indeed, we have an equality in the previous proposition. In order to achieve it, we need some previous results.

One important interpretation of the order of a differential operator is given by the following proposition. It essentially says that a differential operator of order n is determined by its values at polynomials up to degree n .

For each $m < \omega$, we have naturally defined linear maps of k -vector spaces.

$$\Phi_m : \text{End}_k(k[x_1, \dots, x_n]) \rightarrow \text{Hom}_k(k[x_1, \dots, x_n]/\langle x_1, \dots, x_n \rangle^{m+1}, k[x_1, \dots, x_n]).$$

Observe that Φ_n is surjective.

Proposition 2.8. *The restriction of Φ_m to the subspace $D^m(R)$ is injective. Furthermore,*

$$D^m(R) \cong \text{End}_k(k[x_1, \dots, x_n]) / \ker(\Phi_m) \cong \text{Hom}_k(k[x_1, \dots, x_n]/\langle x_1, \dots, x_n \rangle^{m+1}, k[x_1, \dots, x_n]).$$

Proof. Let $P \in \text{End}_k(k[x_1, \dots, x_n])$. It can be proved by induction that for any $l < \omega$ and $x_{i_0}, \dots, x_{i_l} \in \{x_1, \dots, x_n\}$, we have that

$$[[\dots [P, x_{i_0}] \dots], x_{i_{l-1}}], x_{i_l}] = Px_{i_0} \dots x_{i_l} + \sum_{j \in J} p_j Pq_j$$

where J is a finite set and, for every $j \in J$, $p_j \in k[x_1, \dots, x_n]$ and $q_j \in k[x_1, \dots, x_n]_{\leq l}$.

Now, let $P \in D^m(k[x_1, \dots, x_n])$. By Proposition 2.2 (1), we have that $[[\dots [P, x_{i_0}] \dots], x_{i_{l-1}}], x_{i_l}] = 0$ for $l \geq m$. This, together with the previous observation shows that P is determined by its values on polynomials up to degree m , which implies the first assertion of the proposition. $\square$

The previous proposition allows us to see the construction of $D(k[x_1, \dots, x_n])$ under another light. Observe that, for each $m < \omega$, we have natural surjective k -linear morphisms:

$$\varphi_n : \frac{k[x_1, \dots, x_n]}{\langle x_1, \dots, x_n \rangle^{m+2}} \rightarrow \frac{k[x_1, \dots, x_n]}{\langle x_1, \dots, x_n \rangle^{m+1}}.$$

By left-exactness of the contravariant functor $\text{Hom}_k(-, k[x_1, \dots, x_n])$ we get injective k -linear morphism:

$$\varphi_n^* : \text{Hom}_k\left(\frac{k[x_1, \dots, x_n]}{\langle x_1, \dots, x_n \rangle^{m+1}}, k[x_1, \dots, x_n]\right) \rightarrow \text{Hom}_k\left(\frac{k[x_1, \dots, x_n]}{\langle x_1, \dots, x_n \rangle^{m+2}}, k[x_1, \dots, x_n]\right).$$

These injections correspond, under the previous proposition, to the containments $D^m(R) \subset D^{m+1}(R)$. Also, observe that, by considering compositions, the family of morphisms $\{\varphi_m^*\}_{m < \omega}$ induces a directed system of k -vector spaces over the directed set ω . It is easy to prove, using the previous considerations, that

$$D(R) \cong \lim_{\substack{\longrightarrow \\ m < \omega}} \text{Hom}_k\left(\frac{k[x_1, \dots, x_n]}{\langle x_1, \dots, x_n \rangle^{m+1}}, k[x_1, \dots, x_n]\right).$$

Let us return to our goal:

Theorem 2.9. *$D(k[x_1, \dots, x_n]) = A_n(k)$. Furthermore, the order of an operator in $A_n(k)$ coincides with its order as a differential operator.*

Proof. One direction was already proved in Proposition 2.7. For the other direction, let $m < \omega$ and $P \in D^m(R)$. We will construct a differential operator $Q \in A_n(k) \cap D^m(k[x_1, \dots, x_n])$ such that $P(x^\alpha) = Q(x^\alpha)$ for every $\alpha \in \omega^n$ such that $|\alpha| \leq m$. This is equivalent to the property $\Phi_m(P) = \Phi_m(Q)$. Therefore, by Proposition 2.8 we will conclude that $P = Q$, which proves the remaining direction of the equality.

We pass now to the construction of Q . This operator will be obtained by means of a succession of operators $Q_l \in A_n(k) \cap D^l(k[x_1, \dots, x_n])$ inductively defined for $l = 0, \dots, m$ such that $P(x^\alpha) = Q_l(x^\alpha)$ for every $\alpha \in \omega^n$ such that $|\alpha| \leq l$. Then, our desired Q will be Q_m . Let us define $Q_0 = P(1) \in k[x_1, \dots, x_n]$. Then, Q_0 satisfies the desired property since $(P - Q_0)(1) = P(1) - Q_0(1) = P(1) - P(1) = 0$. Let l be a natural number such that $0 < l < m$, and assume that we have defined a $Q_l \in A_n(k) \cap D^l(k[x_1, \dots, x_n])$ such that $P(x^\alpha) = Q_l(x^\alpha)$ for every $\alpha \in \omega^n$ such that $|\alpha| \leq l$. Let us define $Q_{l+1} := \sum_{|\alpha|=l+1} \frac{(P - Q_l)(x^\alpha)}{\alpha!} \partial^\alpha + Q_l$. Then,

clearly $Q_{l+1} \in A_n(k) \cap D^{l+1}(k[x_1, \dots, x_n])$. Now, let $\beta \in \omega^n$ such that $|\beta| \leq l+1$. We have two cases. Assume first that $|\beta| \leq l$. Then, by Lemma 2.4 and the induction hypothesis, we have

$$Q_{l+1}(x^\beta) = \sum_{|\alpha|=l+1} \frac{(P-Q_l)(x^\alpha)}{\alpha!} \partial^\alpha(x^\beta) + Q_l(x^\beta) = \sum_{|\alpha|=l+1} \frac{(P-Q_l)(x^\alpha)}{\alpha!} \alpha! \cdot \delta_{\alpha,\beta} + P(x^\beta) = P(x^\beta).$$

For the second case, suppose that $|\beta| = l+1$. Then, by Lemma 2.4, we have

$$Q_{l+1}(x^\beta) = \sum_{|\alpha|=l+1} \frac{(P-Q_l)(x^\alpha)}{\alpha!} \alpha! \cdot \delta_{\alpha,\beta} + P(x^\beta) = \frac{(P-Q_l)(x^\beta)}{\beta!} \beta! + Q_l(x^\beta) = P(x^\beta).$$

Therefore, have proved that $P(x^\alpha) = Q_{l+1}(x^\alpha)$ for every $\alpha \in \omega^n$ such that $|\alpha| \leq l+1$, which concludes our definition by induction of the operators Q_l . Then, $Q = Q_m$ is the desired operator. Which concludes the proof. Observe also that the order of P as an element of $A_n(k)$ coincides with the one as a differential operator. $\square$

We comment that the proof of the previous theorem is done in (cite Coutinhho) in a different way. There is used a more differential geometric approach, called the method of potentials.

2.3. Properties of the Weyl algebra and some conjectures.

One of the results of this section is that Weyl algebra is a simple ring. Simple rings are very hard to study since one usually relies on ideals to understand rings. One of those tools are filtrations. The order of an operator of $A_n(k)$ allowed us to define the **order filtration** $D^\bullet$, which is a filtration by k -vector spaces.

There is another filtration that will be useful for us, which is called the Bernstein filtration. To define it, we must define the degree of an operator $P \in A_n(k)$. By Proposition 2.5, every such P can be written uniquely in the form

$$P = \sum_{\alpha, \beta \in \omega^n} c_{\alpha, \beta} x^\alpha \partial^\beta,$$

where $c_{\alpha, \beta} \in k$ is non-zero only for a finite number of indices $\alpha, \beta \in \omega^n$.

Definition 2.10. The **degree of** P is the maximum of $|\alpha| + |\beta|$ where $\alpha, \beta \in \omega^n$ are such that $c_{\alpha, \beta} \neq 0$. The order of $P = 0$ is defined to be 0. The degree is notated $\deg(P)$.

For $m < \omega$. let B^m be the k -subspace of $A_n(k)$ consisting of operators of degree less than or equal to m . Then, $B^\bullet$ is a filtration of $A_n(k)$ called the **Bernstein filtration**.

Proposition 2.11. Let $P, Q \in A_n(k)$. Then,

- $\deg(P + Q) \leq \max\{\deg(P), \deg(Q)\}$;
- $\deg(PQ) = \deg(P) + \deg(Q)$;
- $\deg([P, Q]) \leq \deg(P) + \deg(Q) - 2$.

Next, we define the construction of the graded algebra associated to the Bernstein filtration of $A_n(k)$. First, let us define $B^{-1} = \{0\}$, and $gr_m^B(A_n(k)) := B^m / B^{m-1}$ for every $m < \omega$.

Definition 2.12. The **graded algebra associated to the Bernstein filtration of the n -th Weyl algebra** is

$$gr^B(A_n(k)) := \bigoplus_{m < \omega} gr_m^B(A_n(k)).$$

Observe that the algebra operations of $gr^B(A_n(k))$ are indeed well-defined because of Proposition 2.11. This algebra is a commutative version of $A_n(k)$ due to the following proposition.

Proposition 2.13. $gr^B(A_n(k))$ is a graded k -algebra which is isomorphic, as a graded k -algebra, to $k[x_1, \dots, x_n, \xi_1, \dots, \xi_n]$ with the usual grading.

Proof. First, for $m < \omega$, observe that for each $[P] \in gr_m^B(A_n(k))$ there is a unique $P' \in A_n(k)$ such that $[P] = [P']$ and $deg(P') = m$. This allows us to define a bijective map

$$\begin{aligned} \sigma_m : gr_m^B(A_n(k)) &\rightarrow k[x_1, \dots, x_n, \xi_1, \dots, \xi_n]_m \\ x^\alpha \partial^\beta &\mapsto x^\alpha \xi^\beta, \end{aligned}$$

called the **symbol map**. By putting together the symbol maps, we get a bijective map

$$\sigma : gr^B(A_n(k)) \rightarrow k[x_1, \dots, x_n, \xi_1, \dots, \xi_n]$$

This is a k -algebra morphism since for any two operators $P, Q \in A_n(k)$, by Proposition 2.11, we have that $PQ = QP + [P, Q]$, and $deg([P, Q]) \leq deg(P) + deg(Q) - 2$. Therefore, $[P][Q] = [Q][P]$ in $gr^B(A_n(k))$. $\square$

Theorem 2.14. $A_n(k)$ is a left-Noetherian, non-commutative simple domain.

Proof. Let I be a two sided ideal of $A_n(k)$. Let us define $\sigma(I) = \{\sigma_{deg(P)}(P) \mid P \in I\}$. Then, by the Hilbert Basis Theorem $\langle \sigma(I) \rangle = \langle f_1(x, \xi), \dots, f_l(x, \xi) \rangle$. Then, $I = \langle f_1(x, \partial), \dots, f_l(x, \partial) \rangle$. $\square$

Conjecture 1 (Dixmier conjecture). Any endomorphism of the Weyl algebra is an automorphism.

Conjecture 2 (Jacobian conjecture). Any

Theorem 2.15. The Dixmier conjecture implies the Jacobian conjecture.

2.4. The Rees algebra and quantization. The following beautiful observation was done by Arapura in his notes on D -modules.

Definition 2.16. Let $(R, F_\bullet)$ be a filtered, associative, unital k -algebra. The **Rees algebra** of R is defined to be

$$\tilde{R} = \bigoplus_{k < \omega} F_k \hbar^k \subset R[\hbar].$$

Here R is not necessarily commutative.

Proposition 2.17. (1) The natural morphism $k[\hbar] \rightarrow \tilde{R}$ is flat;
(2) $\tilde{R}/\hbar\tilde{R}$ is isomorphic to $gr(R)$.

For $R = A_n(k)$, we have that $\tilde{R}$ is isomorphic to $H_n(k)$. Therefore, we can see $\hbar$ as the reduced Planck constant parameter. When $\hbar = 1$ we obtain the algebra of quantum observables $A_n(k)$, and for $\hbar = 0$ we get the classical algebra of observables $k[x, \xi]$ on the classical phase space $T^*\mathbb{A}_k^n$.

To continue this discussion, read the papers "An algebra of pseudo-differential operators and quantum mechanics in phase space" and "QUANTUM HAMILTONIAN REDUCTION".

2.5. The sheaf of differential operators on a regular k -algebra.

Definition 2.18. Let R be a commutative ring. We say that R is **regular** if every finitely generated R -module has finite projective dimension.

Theorem 2.19. Let R be a regular k -algebra. Then, $D(R)$ is generated by R and $Der_k(R)$ as a k -algebra.

If S is a multiplicative subset of R , we have a monomorphism $\text{Der}_k(R) \hookrightarrow \text{Der}_k(RS^{-1})$.

Theorem 2.20. *Let X be a smooth k -scheme. Then, there exists a unique sheaf $\mathcal{D}_X$ of k -algebras such that for each affine subset U of X we have that $\mathcal{D}_X(U) = D(\mathcal{O}_X(U))$.*

2.6. The sheaf of differential operators on a non-regular k -algebra.

In this section we want to show that theorem (write and cite the previous theorem) may not hold for a non-regular k algebra. For this, we will consider the coordinate ring of an ordinary cusp.

Let

$$R := \frac{k[x, y]}{\langle y^2 - x^3 \rangle}.$$

Proposition 2.21. *$D(R)$ is not generated, as a k -algebra, by R and $\text{Der}_k(R)$.*

Let t be a new indeterminate. Observe that R and $\frac{k[t^2, y]}{\langle y^2 - t^6 \rangle}$ are isomorphic k -algebras. Let $k[t^2, t^3]$ be the k -subalgebra of $k[t]$ generated by t^2 and t^3 , and $\theta : k[t^2, y] \rightarrow k[t^2, t^3]$ the surjective k -morphism defined by $\theta(t^2) = t^2$ and $\theta(y) = t^3$. Then, $\langle y^2 - t^6 \rangle \subset \ker \theta$. Therefore, θ induces an injective k -morphism $\frac{k[t^2, y]}{\langle y^2 - t^6 \rangle} \rightarrow k[t^2, t^3]$ which is clearly surjective. In conclusion, we have an isomorphism of k -algebras

$$\frac{k[x, y]}{\langle y^2 - x^3 \rangle} \rightarrow k[t^2, t^3],$$

which we will also denote by θ , defined by $\theta(x) = t^2$ and $\theta(y) = t^3$.

Observe that θ induces an isomorphism of Poisson k -algebras $\text{End}_k([t^2, t^3]) \cong \text{End}_k(R)$, given by $T \mapsto \theta^{-1} \circ T \circ \theta$, which restricts to an isomorphism of filtered Poisson k -algebras $D(k[t^2, t^3]) \cong D(R)$, i.e., such that $D^n(k[t^2, t^3])$ and $D^n(R)$ are isomorphic k -vector spaces for each $n < \omega$. Therefore, to conclude our assertion about $D(R)$ it is necessary, and sufficient, to prove it for $D(k[t^2, t^3])$.

Lemma 2.22. *$k[t^2, t^3]$, as a k -vector spaces, has $\{1\} \cup \{t^n\}_{n=2}^\infty$ as a k -basis.*

Proof. Since $\{1\} \cup \{t^n\}_{n=2}^\infty$ is k -independent set, and $k[t^2, t^3] \subset \langle \{1\} \cup \{t^n\}_{n=2}^\infty \rangle$, we just need to prove that the reverse inclusion also holds. By definition of subalgebra generated by a set, we have that $1 \in k[t^2, t^3]$. Now, let us consider t^n for $n \geq 2$. If $n \equiv_3 0$, then $t^n \in k[t^2, t^3]$. This implies also that, if $n \equiv_3 2$, then $t^n \in k[t^2, t^3]$. For the last case, suppose that $n \equiv_3 1$. Since $n \geq 2$, the congruence implies that $n \geq 4$. Therefore, $n - 2 \geq 2$ and $n - 2 \equiv_3 2$. By the previous cases, $t^{n-2} \in k[t^2, t^3]$. Which implies $t^n \in k[t^2, t^3]$. This concludes the proof. $\square$

Lemma 2.23. *Every derivation on $k[t_2, t_3]$ can be uniquely extended to a derivation on $k[t]$. Therefore, we have a natural injection $\text{Der}_k(k[t^2, t^3]) \hookrightarrow \text{Der}_k(k[t])$. Under this identification, we have that $\text{Der}_k(k[t^2, t^3]) = \{p(t)\partial_t \mid p(t) \in \langle t \rangle\} \subset k[t]$.*

Proof. Let $D \in \text{Der}_k(k[t^2, t^3])$. First, we assert that t divides $D(t^2)$ as elements of $k[t]$. If it is not the case, $D(t^2) = r$ for some $r \in k^*$. Since D is a derivation, we have that $D(t^4) = 2t^2 D(t^2) = 2rt^2$. Now, observe that $D(t^6) = D(t^2 t^4) = t^2 D(t^4) + t^4 D(t^2) = 3rt^4$. On the other hand, $D(t^6) = D(t^3 t^3) = 2t^3 D(t^3)$. Therefore, $3rt^4 = 2t^3 D(t^3)$ in $k[t]$. Since $k[t]$ is a domain, we have that $D(t^3) = 3rt/2$, which is a contradiction with the observation preceding the Lemma 2.22. Therefore, we can conclude that t divides $D(t^2)$ as elements of $k[t]$.

Then, $D(t^2) = tp(t)$ for an unique $p(t) \in k[t]$. Define $D(t) := p(t)/2$. This gives us the unique extension of D . Therefore, by this unique extension, we can think of D as an element of

$Der_k(k[t])$. By Proposition 2.6, $D = q(t)\partial_t$ for some $q(t) \in k[t]$. We want to conclude that $q(t) \in \langle t \rangle$. If that were not the case, we would have that $q(t) = r \in k^*$. Then $D(t^2) = r\partial_t(t^2) = 2rt$. But this is a contradiction since $2rt \notin k[t^2, t^3]$.

Reciprocally, observe that every derivation of $q(t)\partial_t \in Der_k(k[t])$ with $q(t) \in \langle t \rangle$ belongs to $Der_k(k[t^2, t^3])$ since, by Lemma 2.22, its restriction to $k[t^2, t^3]$ has its range contained in $k[t^2, t^3]$. $\square$

Observe that the previous lemmas easily imply the following proposition.

Proposition 2.24. *$Der_k(k[t^2, t^3])$ is generated as a $Der_k(k[t^2, t^3])$ -module $t\partial_t$ and $t^2\partial_t$.*

Now, let us denote by B the k -subalgebra of $D(k(t))$ generated by $k(t)$ and ∂_t . Observe that every element of $D(k[t]) = A_n(k)$ can be seen as an element of B . That is, we have a injective morphism of filtered poisson k -algebras (for the order filtration) $A_n(k) \hookrightarrow B$. Let C be the k -subalgebra of $D(k[t^2, t^3])$ generated by $k[t^2, t^3]$ and $Der_k(k[t^2, t^3])$. Therefore, by Proposition 2.24, and under the identifications provided by the previous morphism, and by Lemma 2.23, have the inclusions

$$C \subset D(k[t]) = A_n(k) \subset B \subset D(k(t)).$$

We have an easy way to find elements of $D(k(t))$, that under the previous identifications, belong indeed to $D(k[t^2, t^3])$: observe that we have

$$\{P \in D(k(t)) \mid P(k[t^2, t^3]) \subset k[t^2, t^3]\} \subset D(k[t^2, t^3]).$$

For example, using this observation, its easy to prove that

$$\partial_t^2 - 2t^{-1}\partial_t, t\partial_t^2 - \partial_t, \partial_t^3 - 3t^{-1}\partial_t^2 + 3t^{-2}\partial_t \in D(k[t^2, t^3]).$$

Using Proposition 2.24, it is not a hard task to prove that the previous differential operators do not belong to C , which was the aim of this section.

3. THE SHEAF OF DIFFERENTIAL OPERATORS ON A AFFINE SCHEME

In this section, we are going to construct the sheaf of differential operators on $\text{Spec}(A)$.

Since $\text{End}_k(A)$ is a k -module, we can consider the $\mathcal{O}_{\text{Spec}(k)}$ -module $\widehat{\text{End}_k(A)}$ (Hartshorne, Section 2.5). Let us consider the structural morphism $p : \text{Spec}(A) \rightarrow \text{Spec}(k)$. Then

4. D-MODULES: BASIC CONSTRUCTIONS AND SOME EXAMPLES

The objects of study of these notes are the following:

Definition 4.1. Let X be a smooth algebraic variety over a field of characteristic zero, and $\mathcal{D}_X$ its sheaf of differential operators. A left $\mathcal{D}_X$ -module is a sheaf $\mathcal{M}$ on X such that $\mathcal{M}(U)$ is a left $\mathcal{D}_X(U)$ -module for each open subset U of X , and such that these actions are compatible with the restriction homomorphisms.

4.1. The D -module associated to a differential equation.

4.2. The Fourier transform.

4.3. Filtered $\mathcal{D}$ -modules.

5. HOLONOMIC $\mathcal{D}$ -MODULES

5.1. Numerical polynomials. In this section, we consider functions $f : \mathbb{C} \rightarrow \mathbb{C}$ and its discrete variation $\Delta f := f(t+1) - f(t)$. Δf can be thought of as a discrete derivative. When working with discrete variations of polynomials, there is a very useful basis of $k[t]$ which is not so well-known.

Let $r < \omega$. Let us define the binomial coefficients $\binom{t}{r} \in \mathbb{Q}[t]$ in the following way

$$\binom{t}{r} = \begin{cases} \frac{t(t-1)\cdots(t-r+1)}{r!} & \text{if } r > 0, \\ 1 & \text{if } r = 0, \\ 0 & \text{if } r < 0. \end{cases} \quad (5.1)$$

Observe that $\binom{t}{r}$ is a polynomial of degree r . An easy induction on r shows that Pascal's formula also holds in this setting

$$\binom{t+1}{r} = \binom{t}{r-1} + \binom{t}{r}.$$

This tells us that the discrete variation behaves similarly to the derivative $(t^r)' = rt^{r-1}$:

$$\Delta \binom{t}{r} = \binom{t}{r-1}.$$

This property has the following easy but important consequence.

Proposition 5.1. *The binomial coefficients $\binom{t}{r}$, for $r < \omega$, form a k -basis of $k[t]$. Furthermore, for each $n < \omega$, the coefficients $\binom{t}{0}, \binom{t}{1}, \dots, \binom{t}{n}$ form a k -basis of $k[t]_{\leq n}$.*

Proof. To prove the former assertion it is sufficient to prove the later. Let $n < \omega$. Since $\dim_k k[t]_{\leq n} = n+1$, it suffices to prove that $\binom{t}{0}, \binom{t}{1}, \dots, \binom{t}{n}$ are k -linearly independent. Let $c_0, \dots, c_n \in k$ such that

$$\sum_{r=0}^n c_r \binom{t}{r} = 0. \quad (5.2)$$

Let $t = 0$ in 5.2. Then, $c_0 = 0$. Observe that 5.2 implies that

$$\sum_{r=0}^{n-1} c_{r+1} \Delta \binom{t}{r} = \sum_{r=0}^{n-1} c_{r+1} \binom{t}{r} = 0$$

If we let $t = 0$ in the previous equation, we get $c_1 = 0$. In the same way we get $c_2 = \dots = c_n = 0$. $\square$

Definition 5.2. A numerical polynomial is a polynomial $p(t) \in \mathbb{Q}[t]$ such that $p(n) \in \mathbb{Z}$ for integers $n \gg 0$.

Observe that $\mathbb{Z}$ -linear combinations of binomial coefficients are numerical polynomials. The following proposition shows that the converse is also true. It also says that *the value of a numerical polynomial at an integer is always an integer*. Before proving this you should try to do the next exercise.

Exercise 1. For every $n, r \in \mathbb{Z}$, prove that $\binom{n}{r} \in \mathbb{Z}$.

Proposition 5.3. *Let $p(t) \in \mathbb{Q}[t]$ such that $p(n) \in \mathbb{Z}$ for $n \gg 0$. Then, there are integers $c_0, \dots, c_m$ such that*

$$p(t) = \sum_{r=0}^m c_r \binom{t}{r}.$$

In particular, $p(t)$ is a numerical polynomial.

Proof. Let us do a complete induction on $m = \deg(p)$. Suppose that the proposition holds for polynomials of degree less than m , and let $p(t)$ be a polynomial of degree m satisfying the hypotheses of the proposition. By Proposition 5.1, there are unique $c_0, \dots, c_m \in \mathbb{Q}$ such that

$$p(t) = \sum_{r=0}^m c_r \binom{t}{r}$$

. Observe that

$$\Delta p(t) = \sum_{r=0}^{m-1} c_{r+1} \binom{t}{r}$$

also satisfies the hypotheses of the proposition. Therefore, by the induction hypothesis and Proposition 5.1, we have that $c_1, \dots, c_m \in \mathbb{Z}$.

Now, let $n \gg 0$ such that $p(n) \in \mathbb{Z}$. Then,

$$c_0 = p(n) - \sum_{r=1}^m c_r \binom{n}{r} \in \mathbb{Z}.$$

This concludes the proof. $\square$

Proposition 5.4. *Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function. Suppose there is some polynomial $q(t) \in \mathbb{Q}[t]$ of degree d such that $\Delta f(n) = q(n)$ for every $n \gg 0$. Then, there is a polynomial $p(t) \in \mathbb{Q}[t]$ of degree $d + 1$ such that $f(n) = p(n)$ for every $n \gg 0$.*

Proof. Let $N \in \mathbb{N}$ such that $\Delta f(n) = q(n)$ for every $n \geq N$. Let us define $g : \mathbb{N} \rightarrow \mathbb{N}$ by $g(n) = f(n + N)$ for every $n \in \mathbb{N}$. Then, $\Delta g(n) = q(n + N)$ for every $n \in \mathbb{N}$. Observe that, by the hypotheses of the proposition, $q(t)$ is a numerical polynomial of degree d , which implies that $q(t + N)$ is also a numerical polynomial of degree d . Let $c_0, \dots, c_d \in \mathbb{Z}$ such that

$$q(t + N) = \sum_{r=0}^d c_r \binom{t}{r}.$$

Let us define the numerical polynomial $p(t) \in \mathbb{Q}[t]$ of degree $d + 1$ by

$$p(t) = \sum_{r=1}^{d+1} c_{r-1} \binom{t}{r} + g(0).$$

Then, we have that $\Delta p(t) = q(t + N)$. Which implies that $\Delta g(n) = \Delta p(n)$ for every $n \in \mathbb{N}$. This equality, together with $g(0) = p(0)$, implies that $f(n + N) = g(n) = p(n)$ for every $n \in \mathbb{N}$. Then, $f(n) = p(n - N)$ for $n \gg 0$. Observe that, since $p(t)$ is a numerical polynomial of degree $d + 1$, then $p(t - N)$ has also these properties. This concludes the proof. $\square$

5.2. The Hilbert polynomial and the dimension of a module. Let $M = \bigoplus_{i \in \mathbb{Z}} M_i$ be a graded module over $k[x_1, \dots, x_n]$.

Definition 5.5. The **Hilbert function** of M is given by $HF_M(i) = \dim_k M_i$ for every $i \in \mathbb{Z}$.

Theorem 5.6 (Hilbert). *Suppose M is finitely generated over $k[x_1, \dots, x_n]$. Then, there is an unique numerical polynomial $p(t)$, called the **Hilbert polynomial of M** , such that*

$$HF_M(i) = p(i) \quad (5.3)$$

for $i \gg 0$.

Proof. The proof will be done by induction on the number of variables n of the polynomial ring $k[x_1, \dots, x_n]$. For $n = 0$, the hypothesis imply that M is a k -vector space of finite dimension. Therefore, $HF_M(i) = 0$ for $i \gg 0$. Then, we can take $p(t) = 0$.

Suppose the theorem holds for $n \geq 0$, and let M be a graded finitely generated module over $k[x_1, \dots, x_{n+1}]$. First, observe that, since $k[x_1, \dots, x_{n+1}]$ is Noetherian, M is also Noetherian. Let us consider the morphism of $k[x_1, \dots, x_{n+1}]$ -graded modules

$$\mu : M[-1] \rightarrow M$$

given by left multiplication by x_{n+1} . Then, $\ker \mu$ is a graded submodule of $M[-1]$. Since M is Noetherian, there are $m_1, \dots, m_l \in M$ which generate $\ker \mu$ over $k[x_1, \dots, x_{n+1}]$; that is, $\ker \mu = \langle m_1, \dots, m_l \rangle$. Observe that $m_1, \dots, m_l$ also generate $\ker \mu$ over $k[x_1, \dots, x_n]$, since x_{n+1} annihilates $\ker \mu$. Therefore, $\ker \mu$, with the grading inherited from $M[-1]$, is a finitely generated graded module over $k[x_1, \dots, x_n]$. By the induction hypothesis, there is a numerical polynomial $q(t)$ such that $HF_{\ker \mu}(i) = q(i)$ for $i \gg 0$.

Let us consider the following exact sequence of graded $k[x_1, \dots, x_{n+1}]$ -modules

$$0 \rightarrow M[-1] / \ker \mu \xrightarrow{\mu} M \rightarrow \operatorname{coker} \mu \rightarrow 0.$$

Then,

$$HF_M(i) = HF_{M[-1] / \ker \mu}(i) + HF_{\operatorname{coker} \mu}(i) = HF_M(i-1) - HF_{\ker \mu}(i) + HF_{\operatorname{coker} \mu}(i).$$

Therefore, $\Delta HF_M(i-1) = HF_{\operatorname{coker} \mu}(i) - HF_{\ker \mu}(i)$.

Observe that, as M is a finitely generated $k[x_1, \dots, x_{n+1}]$ -module, then $\operatorname{coker} \mu$ is also finitely generated over $k[x_1, \dots, x_{n+1}]$, since it is a quotient of M . Furthermore, since x_{n+1} annihilates $\operatorname{coker} \mu$, then $\operatorname{coker} \mu$ is a graded finitely generated module over $k[x_1, \dots, x_n]$. By the induction hypothesis, there is a numerical polynomial $r(t)$ such that $HF_{\operatorname{coker} \mu}(i) = r(i)$ for $i \gg 0$. Therefore, $\Delta HF_M(i) = HF_{\operatorname{coker} \mu}(i+1) - HF_{\ker \mu}(i+1) = r(i+1) - q(i+1)$ for $i \gg 0$. Since $r(t+1) - q(t+1)$ is a numerical polynomial, Proposition 5.4 allows us to conclude the induction step.

The uniqueness assertion follows by considering the difference of two polynomials satisfying (5.3). $\square$

Definition 5.7. Suppose M is bounded below. The **Hilbert-Samuel function** of M is given by $\chi_M(i) = \sum_{j \leq i} \dim_k M_j$ for every $i \in \mathbb{Z}$.

Corollary 5.8 (Hilbert-Samuel). *Suppose M is finitely generated over $k[x_1, \dots, x_n]$ and bounded below. Then, there is an unique numerical polynomial $q(t)$, called the **Hilbert-Samuel polynomial of M** such that*

$$\chi_M(i) = q(i)$$

for $i \gg 0$.

Proof. The proof follows by observing that $\Delta\chi_M = HF_M$, and applying Theorem 5.6 and Proposition 5.4. $\square$

Proposition 5.9. *Let L, M and N be finitely generated and graded modules over $k[x_1, \dots, x_n]$, and let*

$$0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0 \quad (5.4)$$

be an exact sequence of graded modules over $k[x_1, \dots, x_n]$.

Then,

$$HF_M = HF_L + HF_N. \quad (5.5)$$

The same holds for the Hilbert polynomials. Furthermore, if L, M and N are bounded below, then

$$\chi_M = \chi_L + \chi_N, \quad (5.6)$$

and the same holds for the Hilbert-Samuel polynomials. This, in turn, implies the latter assertions for the bounded below case.

Proof. Since the exact sequence (5.4) is graded, the equality (5.5) follows by considering the induced exact sequence on graded parts. The assertion for the Hilbert polynomials follows from equality (5.5) and the uniqueness of Hilbert polynomials. $\square$

Example 5.10. Let M be a graded and finitely generated module over $k[x_1, \dots, x_n]$, and $m \in M$ a non-torsion, homogeneous element of degree d . We denote by $\langle m \rangle$ the submodule of M generated by m . Then,

$$HF_{\langle m \rangle}(i) = \dim_k(m \cdot k[x_1, \dots, x_n]_{i-d}) = \dim_k(k[x_1, \dots, x_n]_{i-d}) = \binom{i+n-d+1}{n-1}.$$

Therefore, the Hilbert polynomial of $\langle m \rangle$ is

$$p(t) = \binom{t+n-d+1}{n-1} = \frac{t^{n-1}}{(n-1)!} + O(t^{n-1}),$$

and its Hilbert-Samuel polynomial is

$$q(t) = \frac{t^n}{(n-1)!} + O(t^n).$$

5.3. Good filtrations. In this section we will consider filtered $A_n(k)$ -modules for the Bernstein filtration B .

Definition 5.11. A left $A_n(k)$ -module M is said to be **filtered** if there is a family $\Gamma = (\Gamma^i)_{i < \omega}$ of finite dimensional k -vector subspaces of M such that:

- (1) $\Gamma^i \subseteq \Gamma^{i+1}$ for every $i < \omega$;
- (2) $\bigcup_{i < \omega} \Gamma^i = M$;
- (3) $B^i \Gamma^j \subseteq \Gamma^{i+j}$ for every $i, j < \omega$.

We also say that $\Gamma = (\Gamma^i)_{i < \omega}$ is a **(ascending) filtration of M** .

Next, we define the construction of the graded module associated to the filtration Γ . First, let us define $\Gamma^{-1} = \{0\}$, and $gr_i^\Gamma(M) := \Gamma^i / \Gamma^{i-1}$ for every $i < \omega$.

Definition 5.12. The **graded module associated to the filtration Γ** is

$$gr^\Gamma(M) := \bigoplus_{i < \omega} gr_i^\Gamma(M).$$

Observe that by condition (3) of Definition 5.11 it is a graded $gr^B(A_n(k))$ -module. Indeed, let $\sum_{i < \omega} [P_i] \in gr^B(A_n(k))$ where $P_i \in B^i - (B^{i-1} - \{0\})$, and $\sum_{j < \omega} [m_j] \in gr^\Gamma(M)$ here $[m_j] \in gr_j^\Gamma(M)$. Then, multiplication in $gr^\Gamma(M)$ is defined by

$$\left(\sum_{i < \omega} [P_i] \right) \left(\sum_{j < \omega} [m_j] \right) = \sum_{k < \omega} \left(\sum_{i+j=k} [P_i m_j] \right)$$

. Here $\sum_{i+j=k} [P_i m_j] \in gr_k^\Gamma(M)$, and the sum the sum of $gr_j^\Gamma(M)$ as a k -vector space. It is left to reader to prove that this gives well-defined structure of $gr^B(A_n(k))$ -module on $gr^\Gamma(M)$

Proposition 5.13. *Let M be a left $A_n(k)$ -module with filtration Γ , and N be a left $A_n(k)$ -submodule of M . Then, the family of k -vectors spaces $\Gamma \cap N = (\Gamma^i \cap N)_{i < \omega}$ is a filtration of N , and we have a natural injective morphism of $gr^B(A_n(k))$ -modules $gr^{\Gamma \cap N}(N) \rightarrow gr^\Gamma(M)$. By means of this morphism, $gr^{\Gamma \cap N}(N)$ can be seen as a homogeneous $gr^B(A_n(k))$ -submodule of $gr^\Gamma(M)$.*

Proof. It is trivial to verify that $\Gamma \cap N$ indeed defines a filtration of N . Now, observe that, for each $i < \omega$, we have a natural injective k -linear map $gr_i^{\Gamma \cap N}(N) \rightarrow gr_i^\Gamma(M)$ given by $[m] \mapsto [m]$. Putting together these maps gives us the desired injective morphism $gr^{\Gamma \cap N}(N) \rightarrow gr^\Gamma(M)$, which is indeed a $gr^B(A_n(k))$ -modules as a consequence of N being a left $A_n(k)$ -submodule of M . By construction, this morphism is graded. $\square$

Proposition 5.14. *Let M be a left $A_n(k)$ -module with a filtration Γ . If $gr^\Gamma(M)$ is a Noetherian $gr^B(A_n(k))$ -module, then M is Noetherian.*

Proof. Suppose $gr^\Gamma(M)$ is a Noetherian $gr^B(A_n(k))$ -module. Let N be a left $A_n(k)$ -submodule of M . By Proposition 5.13, $gr^{\Gamma \cap N}(N)$ can be seen as a homogeneous $gr^B(A_n(k))$ -submodule of $gr^\Gamma(M)$. Therefore, by hypothesis, there are homogeneous elements $[m_1], \dots, [m_l] \in gr^{\Gamma \cap N}(N)$ such that $gr^{\Gamma \cap N}(N) = \langle [m_1], \dots, [m_l] \rangle$.

Let $m \in N$. If $m = 0$ then $m \in \langle m_1, \dots, m_l \rangle$. Let us suppose $m \neq 0$. Then, we can define $I(m) = \min\{i < \omega \mid m \in \Gamma^i\}$. Observe that $I(m) \geq 0$. Now, we are going to prove, by induction on $I(m)$, that $m \in \langle m_1, \dots, m_l \rangle$. For the base case, suppose $I(m) = 0$. Let us consider $[m] \in gr_0^{\Gamma \cap N}(N)$. COMPLETARRRRR $\square$

Definition 5.15. Let M be a left $A_n(k)$ -module. A **good filtration** of M is a filtration Γ on M such that $gr^\Gamma(M)$ is a finitely generated $gr^B(A_n(k))$ -module.

Let us begin with an example of a non-good (bad) filtration.

Proposition 5.16. *Let M be a left $A_n(k)$ -module with a filtration Γ . Then, Γ is a good filtration if, and only if, exists $k_0 < \omega$ such that $B^i \Gamma^j = \Gamma^{i+j}$ for every $i < \omega$ and $j \geq k_0$.*

Proof. First, suppose that Γ is a good filtration, and let $gr^\Gamma(M)$ be generated by the non-null elements $[m_1], \dots, [m_l]$ as a $gr^B(A_n(k))$ -module. Let $k_0 = \max\{\deg([m_k]) \mid k = 1, \dots, l\}$. Let $j \geq k_0$, $i < \omega$ and $m \in \Gamma^{i+j}$. Then, $[m] \in gr_{i+j}^\Gamma(M)$. Therefore, $[m] = [\sum_k P_k m_k]$, where $P_k m_k \in \Gamma^{i+j}$ for each k . Therefore, $m = \sum_k P_k m_k + m'$ for some $m' \in \Gamma^{i+j-1}$. (The proof must be done by induction on j !!!!). Let $\deg([m_k]) = d_k$. By definition, $m_k \in \Gamma^{d_k}$, but $m_k \notin \Gamma^{d_k-1}$. $\square$

5.4. Bernstein's inequality. jjj

6. SHEAVES

It is important to note that the main objective of sheaves is studying its stalks but not necessarily its sections!!! If we study sections, then we study stalks. But the converse is falsa.

Observe that every flasque sheaf of modules is generated by its global sections. But the converse is false: take any quasicoherent sheaf with non vanishing cohomology.

One thing to prove tomorrow: let $F \rightarrow G$ be a morphism between sheaves of modules that is surjective. Show that under suitable assumptions of F or G being flasque, we have that the image presheaf is indeed a sheaf, and that the coordinates or the morphism are surjective too.

6.1. Morphisms of ringed spaces. First, let us recall the adjunction between the direct and inverse image of sheaves.

Let $f : X \rightarrow Y$ be a continuous map between topological spaces. Then, we have functors

$$\begin{aligned} f_* : \text{AbSh}_X &\rightarrow \text{AbSh}_Y \\ f^{-1} : \text{AbSh}_Y &\rightarrow \text{AbSh}_X \end{aligned}$$

such that $f^{-1} \dashv f_*$. Furthermore, if $\mathcal{F}$ and $\mathcal{G}$ are sheaves on X and Y , respectively, the adjunction gives us a natural bijection

$$\frac{\mathcal{G} \rightarrow f_* \mathcal{F}}{f^{-1} \mathcal{G} \rightarrow \mathcal{F}}$$

Then, for any $x \in X$, the stalk of the top morphism at $f(x)$ (composed with... ADD THE DETAILS!!!) coincides with the stalk of the bottom morphism at x

Therefore, we can define a morphism of ringed spaces equivalently as $(X, \mathcal{O}_X) \xrightarrow{(f, f^\#)} (Y, \mathcal{O}_Y)$ or as $(X, \mathcal{O}_X) \xrightarrow{(f, f^\#)} (Y, \mathcal{O}_Y)$, and obtain the same morphism at the stalks $\mathcal{G}_{f(x)} \rightarrow \mathcal{F}_x$. Then, the following definition is independent of using direct or inverse image for the notion of morphism

Definition 6.1. Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be locally ringed spaces. A **morphism of locally ringed spaces** $(X, \mathcal{O}_X) \xrightarrow{(f, f^\#)} (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces such that, for every $x \in X$, the induced morphism at the stalk $\mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$ is local.

6.2. Quasi-coherent modules.

Proposition 6.2. Let $X = \text{Spec}(A)$ be an affine scheme and $\mathcal{F}$ an $\mathcal{O}_X$ -module. The following assertions are equivalent:

- (1) $\mathcal{F}$ is of the form $\widetilde{M}$ for some A -module M ;
- (2) There are sets I and J , and an exact sequence of $\mathcal{O}_X$ -modules

$$\mathcal{O}_X^{(I)} \rightarrow \mathcal{O}_X^{(J)} \rightarrow \mathcal{F} \rightarrow 0;$$

- (3) There are A -modules M and N , and an exact sequence of $\mathcal{O}_X$ -modules

$$\widetilde{M} \rightarrow \widetilde{N} \rightarrow \mathcal{F} \rightarrow 0;$$

- (4) For each $f \in A$, the natural morphism of A_f -modules

$$\mathcal{F}(\text{Spec}(A)) \otimes_A A_f \rightarrow \mathcal{F}(D(f))$$

is an isomorphism.

Proof. Since the natural morphisms of modules in (4) define a morphism $\widetilde{\mathcal{F}(\text{Spec}(A))} \rightarrow \mathcal{F}$ of $\mathcal{O}_X$ -modules, it is evident that (1) and (4) are equivalent.

Let us prove now that (2) $\Rightarrow$ (4). We begin by proving the desired isomorphisms at the level of stalks. That is, let $\mathfrak{p} \in \text{Spec}(A)$. We are going to prove that the morphism of $A_{\mathfrak{p}}$ -modules

$$\Theta_{\mathfrak{p}} : \mathcal{F}(\text{Spec}(A)) \otimes_A A_{\mathfrak{p}} \rightarrow \mathcal{F}_{\mathfrak{p}}$$

is an isomorphism.

By hypothesis, we have an exact sequence $\mathcal{O}_X^{(I)} \xrightarrow{\psi} \mathcal{O}_X^{(J)} \xrightarrow{\phi} \mathcal{F} \rightarrow 0$. Let us denote by $\{\delta_i \in A^{(I)}\}_{i \in I}$ and $\{\epsilon_j \in A^{(J)}\}_{j \in J}$ the corresponding bases of $A^{(I)}$ and $A^{(J)}$, respectively. By exactness of the previous sequence, we have an exact sequence of $A_{\mathfrak{p}}$ -modules

$$A_{\mathfrak{p}}^{(I)} \xrightarrow{\psi} A_{\mathfrak{p}}^{(J)} \xrightarrow{\phi} \mathcal{F}_{\mathfrak{p}} \rightarrow 0.$$

Let $m \in \mathcal{F}_{\mathfrak{p}}$. Then, there are $a_j \in A$ and $b \in A - \mathfrak{p}$, such that $m = \frac{1}{b} \sum_{j \in J} a_j \phi(\epsilon_j)$. This implies that $\Theta_{\mathfrak{p}}((\sum_{j \in J} a_j \phi(\epsilon_j)) \otimes \frac{1}{b}) = m$. Therefore, $\Theta_{\mathfrak{p}}$ is surjective. To check injectivity of $\Theta_{\mathfrak{p}}$, let $m \otimes \frac{1}{b} \in \mathcal{F}(\text{Spec}(A)) \otimes_A A_{\mathfrak{p}}$ be such that $\Theta_{\mathfrak{p}}(m \otimes \frac{1}{b}) = \frac{1}{b} m = 0$ in $\mathcal{F}_{\mathfrak{p}}$. Then, by exactness of the previous sequence, there is $m' \in A_{\mathfrak{p}}^{(J)}$, such that $\phi(m') = \frac{1}{b} m = 0$. Again, by exactness of the previous sequence, there are $c_i \in A$ and $d \in A - \mathfrak{p}$ such that $\psi(\frac{1}{d} \sum_{i \in I} c_i \delta_i) = m'$. Therefore $(\phi \circ \psi)(\frac{1}{d} \sum_{i \in I} c_i \delta_i) = dm$ in $\mathcal{F}(\text{Spec}(A))$. Now, by exactness of $\mathcal{O}_X^{(I)} \xrightarrow{\psi} \mathcal{O}_X^{(J)} \xrightarrow{\phi} \mathcal{F} \rightarrow 0$, we have that (prove this in detail) $\phi \circ \psi = 0$ in the (not necessarily exact) global sections sequence

$$A^{(I)} \xrightarrow{\psi} A^{(J)} \xrightarrow{\phi} \mathcal{F}(\text{Spec}(A)) \rightarrow 0.$$

Therefore, $dm = 0$ in $\mathcal{F}(\text{Spec}(A))$. Which implies $m \otimes \frac{1}{b} = m \otimes \frac{d}{bd} = dm \otimes \frac{1}{bd} = 0$ in $\mathcal{F}(\text{Spec}(A)) \otimes_A A_{\mathfrak{p}}$. Therefore, $\Theta_{\mathfrak{p}}$ is an isomorphism.

Now, let $f \in A$. With the previous result in hand, we are now able to prove that the natural morphism of A_f -modules

$$\Theta_f : \mathcal{F}(\text{Spec}(A)) \otimes_A A_f \rightarrow \mathcal{F}(D(f))$$

is an isomorphism. Let us consider the $\mathcal{O}_X$ -module $\widetilde{F(\text{Spec}(A))}$. Observe that the morphisms Θ_f define a morphism $\Theta : \widetilde{F(\text{Spec}(A))} \rightarrow \mathcal{F}$, and, by our previous result, it is an isomorphism. Therefore, each Θ_f is an isomorphism. □

COMPLETAR LA ANTERIOR PRUEBA!!!!

Definition 6.3. An $\mathcal{O}_X$ -module satisfying any of the conditions of the previous proposition is called a **quasi-coherent** sheaf on X .

6.3. Smooth, unramified and étale morphisms.

Definition 6.4. A closed immersion $T \hookrightarrow T_n$ is called a **thickening of order n** of T if its defining vanishing ideal $\mathcal{I} \subset \mathcal{O}_{T_n}$ satisfies $\mathcal{I}^n = 0$. The thickening $T \hookrightarrow T_n$ is called **affine** if both T and T_n are affine schemes.

Proposition 6.5. Every thickening $T \hookrightarrow T_n$ of order n is indeed a homeomorphism.

Proof. It is enough to prove that the underlying continuous function in $T \hookrightarrow T_n$ is a surjection. Let $p \in T_n$ and $\text{Spec}(A) = U \subset T_n$ an open affine neighborhood of p . Then $X \cap U \hookrightarrow U$ is also a closed immersion. Therefore, $X \cap U = \text{Spec}(A/I)$ for some ideal I such that $I^n = 0$.

Let $\mathfrak{p} \in \operatorname{Spec}(A)$. Then, $I^n = 0 \subset \mathfrak{p}$, which implies $I \subset \mathfrak{p}$. Therefore, $\mathfrak{p} \in \operatorname{Spec}(A/I)$. This completes the proof. $\square$

Definition 6.6. A morphism of schemes $\pi : X \rightarrow S$ is called **smooth** (resp. **unramified**, **étale**) if

- (1) π is locally of finite presentation;
- (2) for every S -scheme T_1 and thickening $T \hookrightarrow T_1$ of order one, the induced map $\operatorname{Hom}_S(T_1, X) \rightarrow \operatorname{Hom}_S(T, X)$ is surjective (resp. injective, bijective).

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